

BOMI KWON

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Research Interests: Diffusion-based structured generation and controllable generative modeling.

EDUCATION

Sookmyung Women's University

Seoul, Republic of Korea

Master of Science, Department of Computer Science

Mar. 2025 – Feb. 2027 (expected)

Bachelor of Science, Department of Statistics

Mar. 2020 – Feb. 2025

Bachelor of Business Administration, Department of Business Administration

PUBLICATIONS & PRESENTATIONS

B. Kwon and K. Y. Lee (2026). *Text-Conditioned Scene Graph Sequence Generation via Discrete Diffusion*. Proceedings of the 35th ACM International Conference on Information and Knowledge Management (CIKM '26). Accepted Short Paper. [Paper]

B. Kwon and K. Y. Lee (2026). *Text-driven 2D Human Pose Generation via Diffusion in Directional Angle Space*. The Transactions of the Korea Information Processing Society, 15(7), 610–619. [Paper]

B. Kwon, S. Park, Y. Park, and K. Y. Lee (2026). *Text-Conditioned Diffusion for Static Humanoid Robot Pose Generation with Joint-Limit Embedding and Self-Collision Guidance*. Submitted to KICS 2026.

B. Kwon and K. Y. Lee (2026). *Diffusion-based Skeleton Pose Editing*. 2026 Spring Conference of the Korean Institute of Smart Media and the Korean Society of Digital Industry. [Paper]

B. Kwon and K. Y. Lee (2025). *A Graph Generation Model for Text-based Graph Editing*. Korea Software Congress 2025 (KSC 2025). [Paper]

B. Kwon and K. Y. Lee (2025). *Development of a Large-Scale Graph Retrieval System based on LLM and Graph Database Providing a Natural Language Interface*. Korea Computer Congress 2025 (KCC 2025). [Paper]

HONORS & AWARDS

Principal Investigator, 2026 STEM Research Team Project

Apr. 2026 – Oct. 2026

Korea Foundation for Women in Science, Engineering and Technology (WISET) [Official News]

- Selected as one of 50 research teams nationwide; awarded KRW 7,000,000 in competitive research funding.

Sookmyung Leadership Activity Scholarship

Jan. 2023

RESEARCH EXPERIENCE

Sookmyung Women's University

Seoul, Republic of Korea

Graduate Researcher, Data Intelligence Laboratory

Mar. 2025 – Present

Advisor: Prof. Ki Yong Lee

WISET

Principal Investigator, STEM Research Team Project

Apr. 2026 – Oct. 2026

- Lead a team of two undergraduate researchers developing synthetic human-pose generation methods for industrial safety.
- Manage budget execution and reporting for the funded research project.

PROJECTS

Text-Conditioned Scene Graph Sequence Generation via Discrete Diffusion

2025 – 2026

- Developed a discrete diffusion framework for generating time-ordered scene graph sequences from natural-language descriptions of human–object interactions.
- Designed a Transformer-based model for modeling within-frame object relations and temporal transitions; constructed training data from Charades-STA and Action Genome and evaluated against retrieval and generative baselines.
- Led the project as first author from problem formulation and model development through experiments and manuscript preparation.

Text-Conditioned Humanoid Robot Pose Generation with Physical Constraints

2026

- Developed a text-conditioned diffusion model for generating static poses for the Unitree G1 humanoid under joint-range and self-collision constraints.
- Retargeted human motion data to Unitree G1 in MuJoCo and incorporated physical constraints into the generation process while preserving text-based control and pose diversity.

Text-driven 2D Human Pose Generation via Diffusion in Directional Angle Space

2025 – 2026

- Proposed a directional-angle representation that models body segments by their directions rather than Cartesian joint coordinates.
- Built a BGE-M3-conditioned diffusion model for pose generation while preserving consistent skeletal proportions.

AISUM Industry–Academic Collaboration — Product Matching and Retrieval System

2025 – Present

- Developed product retrieval pipelines spanning visual product matching and semantic recommendation for large-scale product data.
- Evaluated visual embedding models and ranking strategies, and built semantic retrieval using BGE-m3-ko/KURE-v1 embeddings and Qdrant.
- Integrated the retrieval functions into an MCP-based interface and implemented a LangGraph agent workflow for multi-stage product search.

Large-Scale Graph Retrieval System with LLM and Neo4j

2025

- Developed an LLM-based natural-language interface that converts user queries into Cypher for retrieval over large-scale graph data in Neo4j.

ENTREPRENEURIAL EXPERIENCE

fitapat (ZOOC)

Founding Team Member / Product & Marketing Lead

Jul. 2023 – Aug. 2024

- Co-founded and led product development for an AI-based pet customization service that generated personalized characters from user-provided photos and connected them to customized merchandise.
- Led product and go-to-market execution through launch; the platform and physical products were showcased at CES 2024.

LEADERSHIP & ACTIVITIES

SOPT

Vice President, 32nd Cohort

Jan. 2023 – Jul. 2023

Product Manager, 31st Cohort

Jul. 2022 – Jan. 2023

- Led a nationwide university IT and startup community; previously served as Product Manager for ZOOC, a digital service for pet owners.

Sookmyung Admissions Ambassador “Polaris” — Vice President

Mar. 2021 – Feb. 2023

Statistics Korea (KOSTAT) — Volunteer Statistics Educator

May 2021 – Dec. 2021

Amorepacific × Sookmyung A.I. Team Project — Team Project

Oct. 2024 – Dec. 2024

SOOK-TAT — Statistics Data Analysis Society, Member

Sep. 2022 – Jul. 2023

Department of Statistics Student Council — Member

Mar. 2020 – Dec. 2021

KB Campus Star — Student Ambassador

May 2022 – Nov. 2022

TECHNICAL SKILLS

Programming: Python, R, SQL

ML & AI: PyTorch, Diffusion Models, Transformers, LangChain, LangGraph, MCP

Databases & Retrieval: Qdrant, PostgreSQL/pgvector, Neo4j

Simulation: MuJoCo

Tools: Git, GitHub

Certification: ADsP